



Comprehensive Study Guide for Statistics and Probability Concepts

Statistics and Probability (Universiti Teknologi Malaysia)



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Comprehensive Study Guide on Statistics and Probability

Probability Mass Function and Cumulative Distribution Function

Understanding the probability mass function (PMF) for discrete random variables is fundamental in probability theory. The PMF, denoted as $f(x)$, gives the probability that a discrete random variable (X) takes a specific value (x). For example, in the context of the number of people entering a gym within an hour, the PMF assigns probabilities to each possible count (0, 1, 2, 3, ...).

In the given scenario, the probability function is expressed as: $f(x) = \frac{2}{x+p} \quad \text{for } x = 1, 2, 3, 4, 5$ where (p) is an unknown constant. To find (p) , the key step involves using the fact that the sum of all probabilities in a PMF must equal 1: $\sum_{x=1}^5 f(x) = 1$ Substituting the expression for $f(x)$: $\sum_{x=1}^5 \frac{2}{x+p} = 1$ Solving this equation yields the value of (p) . Once (p) is known, a cumulative distribution function (CDF), denoted as $F(x) = P(X \leq x)$, can be constructed by summing the probabilities up to a certain value: $F(x) = \sum_{k=1}^x f(k)$ The CDF is a non-decreasing step function that provides the probability that (X) is less than or equal to a particular value. Graphically, it is a staircase that jumps at each point where (X) can take a value, approaching 1 as (x) approaches the maximum value.

Probability Density Function of Continuous Random Variables

For continuous random variables, the probability is described by a probability density function (PDF), $f(w)$. Unlike discrete cases, the

probability that (W) takes an exact value is zero; instead, probabilities are assigned over intervals: $[P(a \leq W \leq b) = \int_a^b f(w) , dw]$ In the provided problem, the PDF is defined as: $[f(w) = \begin{cases} k w, & 0 \leq w \leq 1 \\ 0, & \text{otherwise} \end{cases}]$ To ensure $(f(w))$ is a valid PDF, it must satisfy: $[\int_{-\infty}^{\infty} f(w) , dw = 1]$ which reduces to: $[\int_0^1 k w , dw = 1]$ Calculating the integral: $[k \int_0^1 w , dw = k \times \frac{1}{2} = 1]$ solving for (k) : $[k = 2]$ However, the problem states $(k=3)$, indicating a correction or additional constraints in the original context. The key takeaway is that the constant (k) ensures the total probability over the interval sums to 1, making $(f(w))$ a valid density function.

Probabilities over intervals, such as $(P(0 < W < 1.5))$, are computed via integration: $[P(0 < W < 1.5) = \int_0^{1.5} 3w , dw]$ since $(f(w)=0)$ for $(w > 1)$, the probability for $(w > 1)$ is zero, and the integral simplifies accordingly.

Expected Value and Variance of Random Variables

The **expected value** $(E[X])$ (or mean) of a random variable provides the long-term average outcome. For discrete variables: $[E[X] = \sum_{\{x\}} x \times P(X=x)]$ For continuous variables: $[E[X] = \int_{-\infty}^{\infty} x \times f(x) , dx]$ Similarly, the **variance** $(\text{Var}(X))$ measures the spread of the distribution: $[\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2]$ where: $[E[X^2] = \sum_{\{x\}} x^2 P(X=x) \quad \text{(discrete)}]$ or $[E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) , dx \quad \text{(continuous)}]$

In the specific problems, calculating $(E[X])$ involves summing or integrating the product of (x) and its probability density or mass, while $(\text{Var}(X))$ involves computing $(E[X^2])$ and applying the variance formula.

Transformations such as $(Y = c + X)$ alter the expectation and variance:

- $(E[Y] = c + E[X])$
- $(\text{Var}(Y) = \text{Var}(X))$

This is crucial when dealing with linear transformations of random variables, especially in problems involving shifting or scaling.

Binomial and Poisson Distributions in Real-World Contexts

Binomial Distribution:

Models the number of successes in a fixed number of independent Bernoulli trials, each with probability (p) of success. Its probability mass function is: $[P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}]$ where:

- (n) is the number of trials,
- (x) is the number of successes,
- (p) is the probability of success in each trial.

For example, in defect rate scenarios, if the probability of a single component being defective is (p), then the probability that a component is defective is (p), and the total number of defective components in a batch follows a binomial distribution.

Poisson Distribution:

Models the number of events occurring in a fixed interval or space, especially when events happen independently at a constant average rate (λ). Its probability mass function is: $[P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}]$ used when counting rare events, such as the number of people entering a gym in an hour or the number of defects in a batch.

Application:

- The probability that a product contains no defective circuits can be modeled with the Poisson distribution, especially when the average number of defects is known.

- The probability of exactly (k) events (e.g., visitors) in a given time frame can be calculated using the Poisson distribution, given the average rate.

Probability Rules and Calculations for Multiple Events

Various probability rules help in calculating the likelihood of complex events:

- **At least** (k) successes: $P(X \geq k) = 1 - P(X < k) = 1 - P(X \leq k-1)$
- **More than** a certain number: $P(X > x) = 1 - P(X \leq x)$
- **Between** two values: $P(a \leq X \leq b) = P(X \leq b) - P(X < a)$

In batch and quality control scenarios:

- The probability that a batch contains at least a certain number of non-defective items involves summing binomial probabilities.
- The probability that more than a certain number of batches meet quality criteria can be modeled using binomial or normal approximations, especially when the number of batches is large.

Exam Strategies and Academic Integrity

Success in statistical problem-solving requires:

- **Clear, step-by-step solutions:** Show all calculations to facilitate understanding and partial credit.
- **Individual effort:** Avoid sharing answers to maintain fairness and integrity.
- **Understanding key concepts:** Focus on grasping the underlying principles rather than rote memorization.
- **Adherence to rules:** Violations such as copying can lead to disciplinary actions, including expulsion.

- **Effective time management:** Allocate time wisely during exams to ensure all questions are attempted with proper care.

Maintaining academic integrity and demonstrating genuine understanding are essential for mastering statistics and probability.

This guide has synthesized the key concepts, formulas, and problem-solving approaches from the provided materials, offering a comprehensive resource for mastering the subject. # Comprehensive Study Guide on Statistics and Probability

Probability Mass Function and Cumulative Distribution Function

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